Algorithms 2

Matching in graphs

Daniel Rosenberg & Michael Trushkin

Ariel University

1. Tutte's Theorem

1.1 Perfect Matchings

For bipartite graphs $G := (A \cup B, E)$

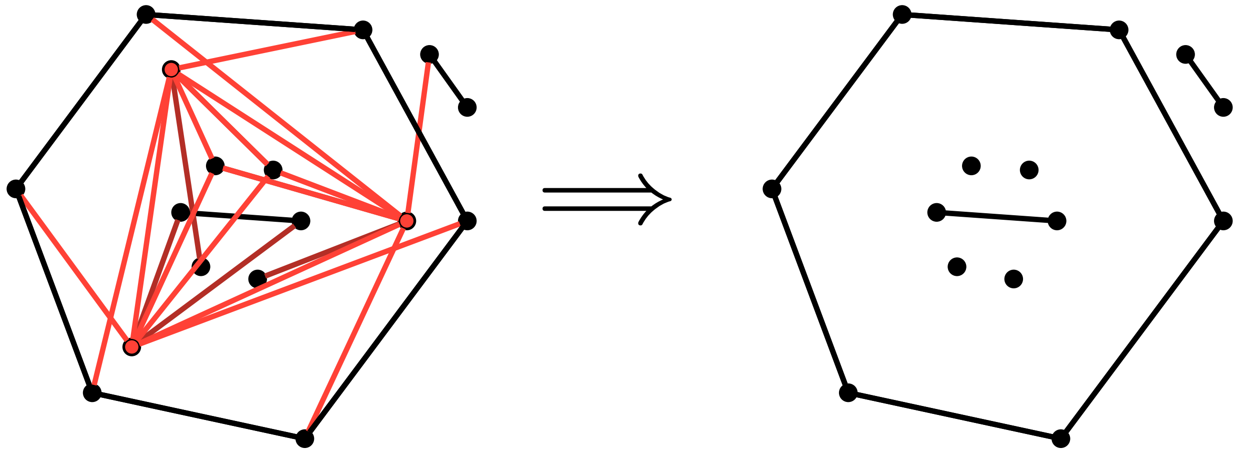
G has a perfect matching $\iff$ $|A| = |B|$ and A satisfoes the Hall-Condition

Question 1.1.1 what if G is not bipartite?

Goal : Find a necessary & sufficient condition for the emergence of perfect matchings in *general* graphs.

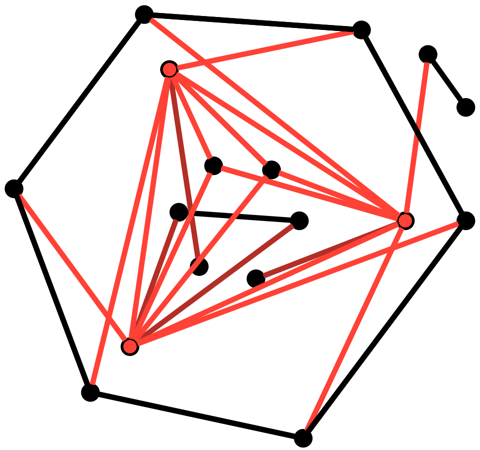
1.2 Removing vertices from graphs

- G graph and $S \subseteq V(G)$
- $G - S :=$ the graph obtained by removing the vertices S in G.
  - $G - S$ is a sub-graph of G.

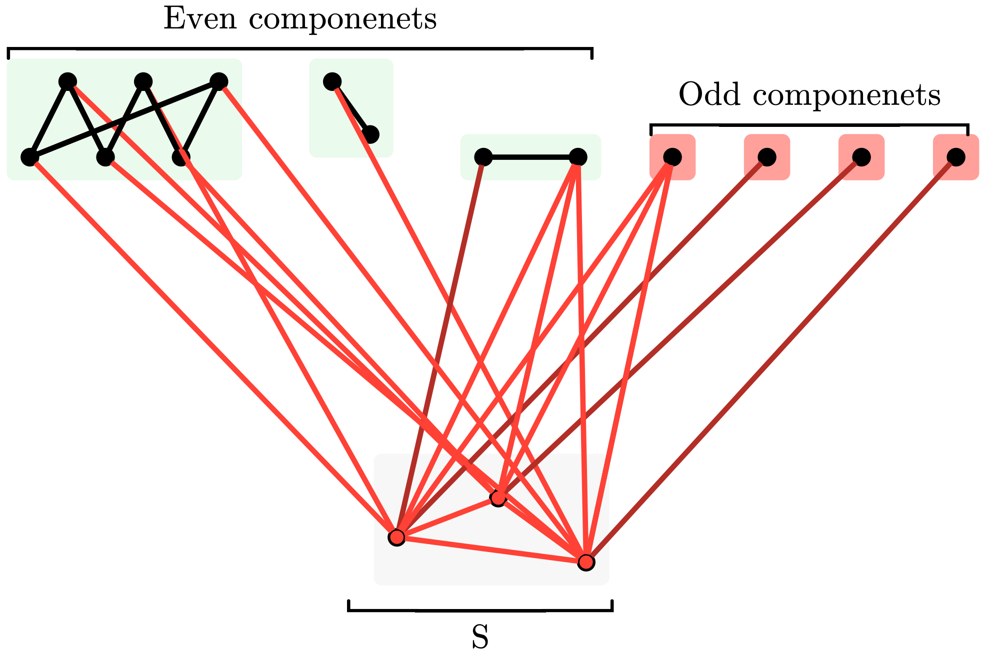


1.3 Removing vertices from graphs

- We can break the graph into the set S and the connected components of $G - S$.
- **No edges between connected componenets**



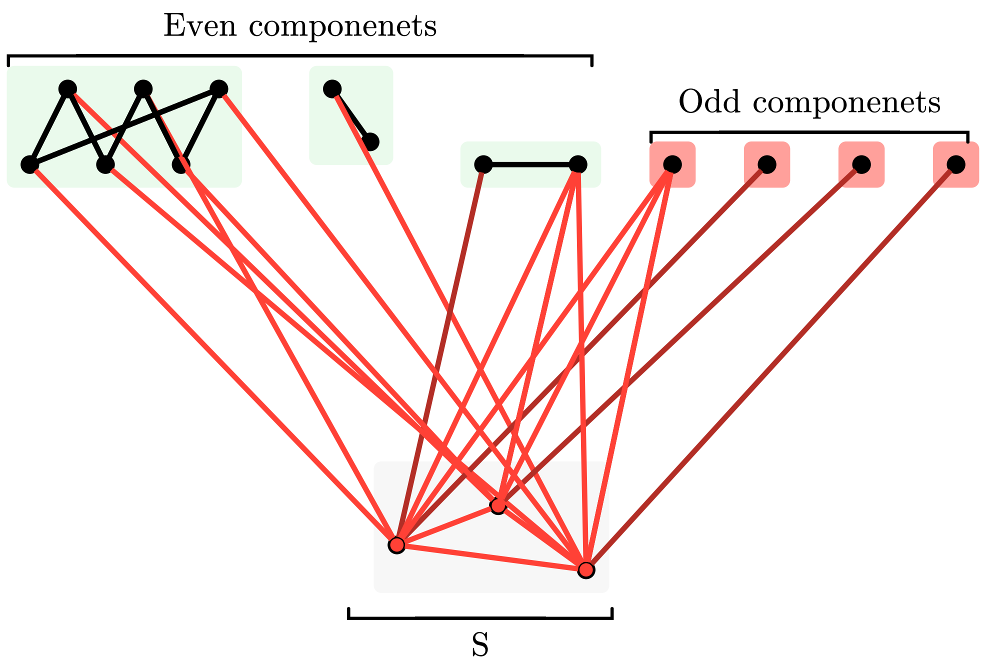
$\Longrightarrow$



No perfect matching, there is 4 componenets, with odd number of vertices.
But S can only "give" at most 3 vertices.

1.4 Conclusion

- Let G be a graph.
- $\mathrm{Co}(G - S) :=$ number of odd components in $G - S$.
- If $\exists S \subseteq V(G)$ s.t. $\mathrm{Co}(G - S) > |S|$
  - G has no perfect matching



1.5 Tutte's Theorem

- $\text{Co}(G - S) :=$ number of odd components in $G - S$.

Theorem 1.5.1 (Tutte's) Let be a graph. Then,

$$G \text{ has a p.m} \iff \underbrace{|\text{Co}(G - S)| \le |S|, \forall S \subseteq V(G))}_{\text{Tutte's Condition}}$$

We already proven $\Longrightarrow$

Remark

A graph satisfying

$$|\text{Co}(G - S)| \le |S|, \forall S \subseteq A.$$

Is said to satisfy the Tutte's condition.

2. General Proof Idea

2.1 General Proof Idea

- Suppose the claim is false.
  - G satisfies Tutte's condition and has no perfect matching.
- Let G be "edge-maximal" example of such a graph.
  - **What does G looks like? What are the properties of such a graph?**

2.2 Edge Maxmimal Counter Example

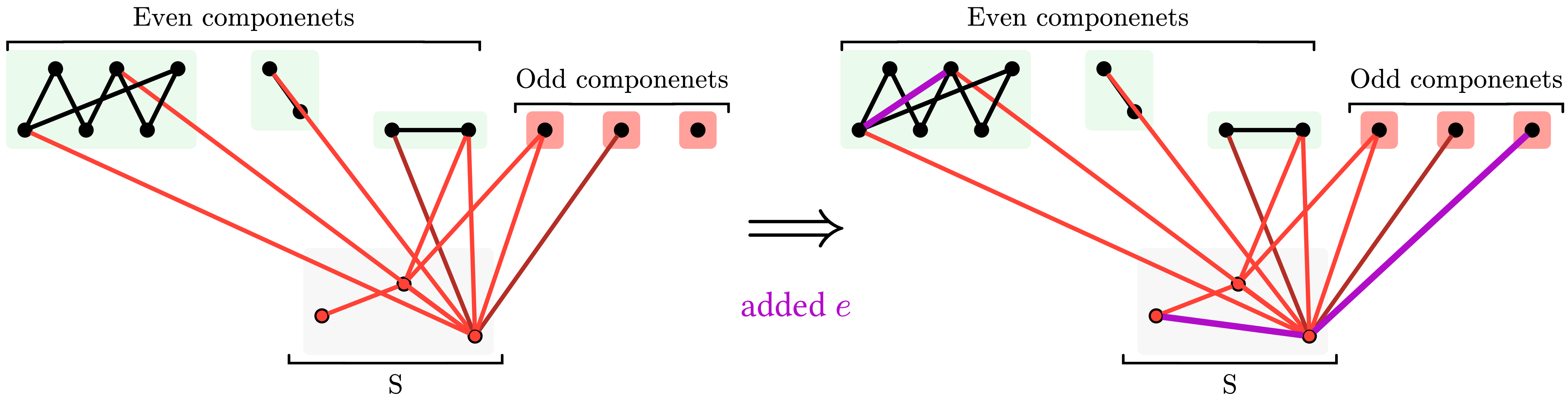
Observation 2.2.1

- G satisfying the tutte condition
- e an edge not in G

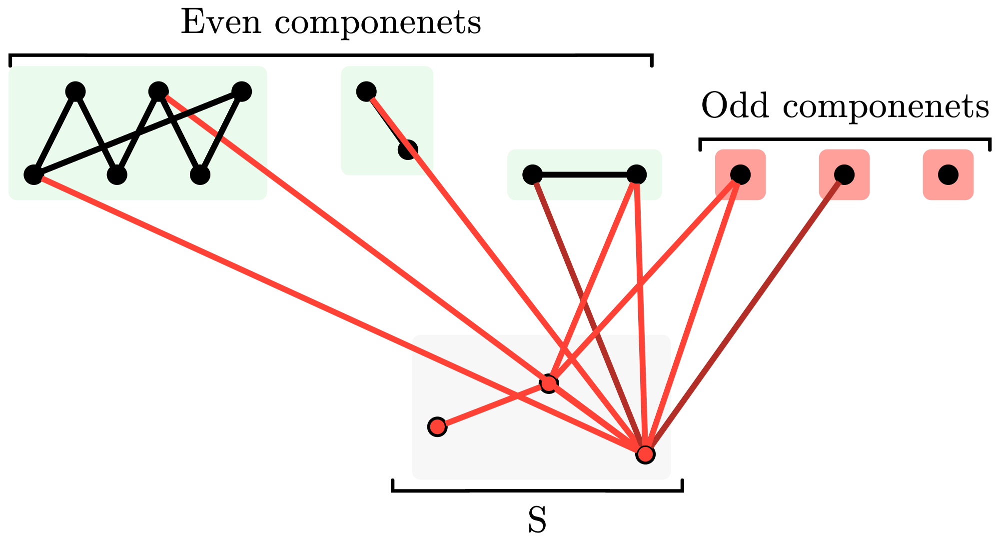
Then, $G' := G + e$ also satisfies the tutte's condition.

Proof. Fix $S \subseteq V(G)$ and show that for any edge $\mathrm{Co}(G - S + e) \leq |S|$.

- if e is an edge that resides inside any component. inside of S, or between S and any component.
  - Then, there is no change $\mathrm{Co}(G - S + e) = \mathrm{Co}(G - S) \leq |S|$.

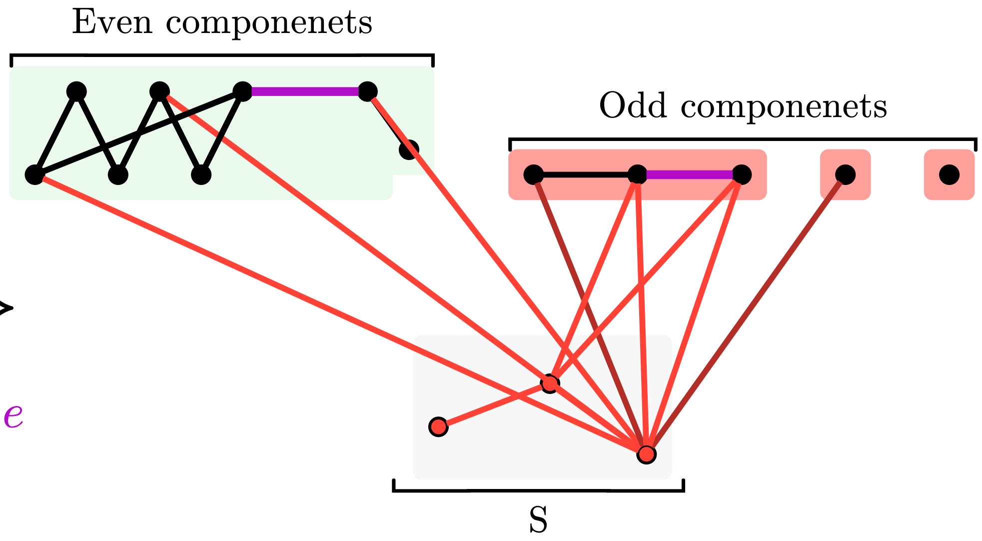


- e is an edge with one end inside an even component:
  - If its between two *even componenets*. Then, we get one big even components instead of two even components so that $\mathrm{Co}(G - S + e) = \mathrm{Co}(G - S) \leq |S|$.
  - If its between *even and odd components*. Then, we get one big odd components instead of the previous odd and even components so that $\mathrm{Co}(G - S + e) = \mathrm{Co}(G - S) \leq |S|$.

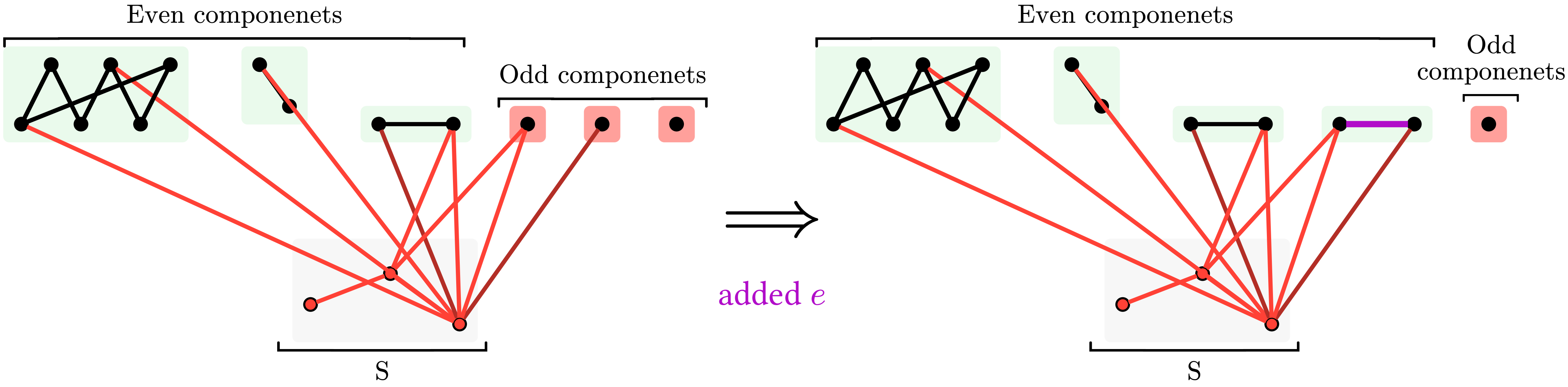


$\Longrightarrow$

added e



- If e is an edge with both ends in different odd components:
  - Then, we get one even component instead of the two odd components so that $\mathrm{Co}(G - S + e) = \mathrm{Co}(G - S) - 2 \leq |S| - 2 \leq |S|$.



Observation 2.3.1

- G graph, satisfying the Tutte's condition

Then, $v(G) = n$ is even.

Tutte's Condition:

$$\mathrm{Co}(G - S)| \leq |S|, \forall S \subseteq A.$$

Proof. Set $S = \emptyset$.

- $\mathrm{Co}(G - S) = \mathrm{Co}(G) \leq |S| = 0$
  - Every connect component of G must have an even number of vertices.

$\square$

3. Construction edge maxinal counter example

3.1 Construction edge maxinal counter example

Theorem 3.1.1 (Tutte's) Let be a graph. Then,

$$G \text{ has a p.m} \quad \Longleftrightarrow \quad \underbrace{|\mathrm{Co}(G-S)| \leq |S|, \forall S \subseteq V(G))}_{\text{Tutte's Condition}}$$

- Assume that $\exists G$ satisfying the Tutte's condition without a p.m.
  - If $\exists e \notin E(G)$ s.t. $G + e$ still has no p.m.
    - Set $G := G + e$.

We obtain a graph G with the following properties:

1. G satisfies the Tutte's condition
2. G has no perfect matching
3. $\forall e \notin E(G) : G + e$ has a perfect matching.

G is called an edge maximal counter example!

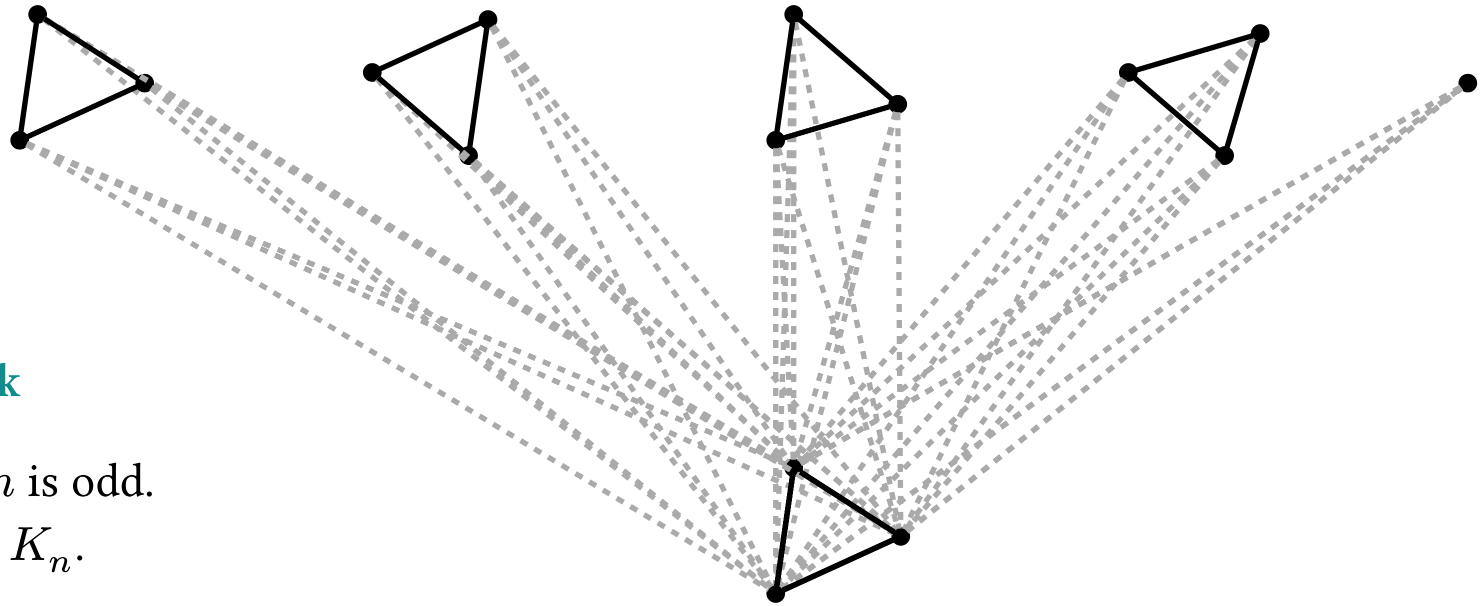
A graph satisfying (2,3) is an **SNF graph!**

SNF graphs do not satisfy Tutte's condition

4. SNF graphs

Definition 4.1.1 (SNF Graph's) A graph G is called **Saturated non-fractorisable** graph if it satisfies:

1. G has no perfect matching
2. $G + e$ has a perfect matching $\forall e \notin E(G)$



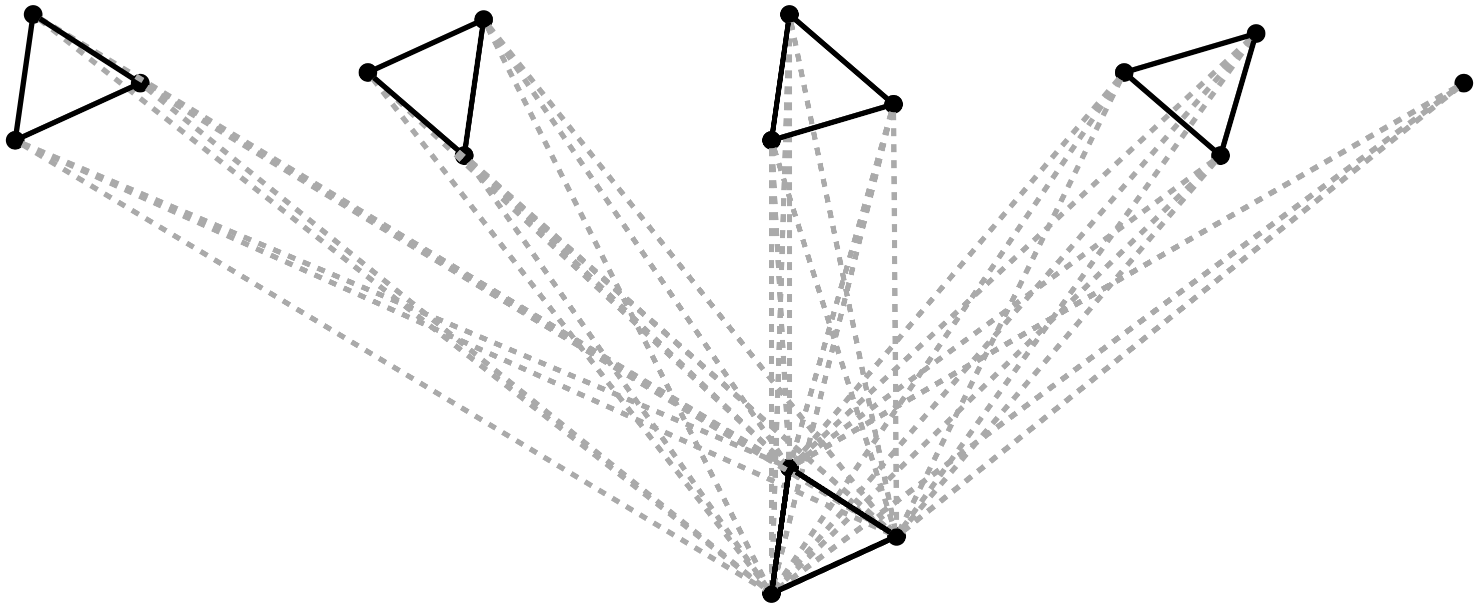
Remark

If $v(G) = n$ is odd.
Then, $G = K_n$.

How do even SNF graph look like? Proof is left for the TA session.

- $\exists S \in V(G)$ and $k \in \mathbb{N}$ s.t. $S = K_k$.
  - $G - S$ consist of $k + 2$ odd cliques.

This means that any SNF graph by construction doesnt sayfies the Tutte's condition! In particular setting S to the K_k we have $\mathrm{Co}(G - S) = k + 2 > k = |S|$.



In the example above, $S = K_3$ and there is 5 odd components in $G - S$.

4.2 Conclusion

Conclusion.

- If G is an even sized **SNF graph**.
  - $\exists S \in V(G)$ s.t. $\mathrm{Co}(G - S) = |S| + 2 > |S|$.

  $\implies G$ doesn't satisfy the Tutte's condition!

Back to the proof: Let G be a counter example i.e.

- G satisfies the Tutte's condition & G has no perfect matching

$\implies$ Construct an edge maximal counter example from G called G^*. $\implies G^*$ is an **SNF graph** & G^* satifies the tutte's condition. $\implies$ Any **SNF graph** doesn't satisfies tutte's so its a contradiction. $\implies G^*$ cannot exists $\implies G$ cannot exists. $\implies$ If G satisfies the Tutte's condition then G must have a perfect matching!